Joshua A. Bisdorf

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Education

The University of Texas at Austin Austin, TX
Master of Science, Computer Science *August 2026 – Expected May 2028*

University of Michigan Ann Arbor, MI
Bachelor of Science in Engineering, Computer Science | *GPA: 3.67* *August 2022 – December 2025*

- Magna Cum Laude

Experience

KBR, Inc. Ann Arbor, MI
Software Engineer | *C++, CUDA, MATLAB, Python* *January 2026 – Present*

- Develop and integrate mission-critical radar software for defense and intelligence systems across signal processing, machine learning, simulation, and computer vision
- Optimize radar application performance by streamlining data-processing, software-engineering, and algorithmic workflows in an agile development environment

Software Engineer Intern | *MATLAB, Python* *May 2024 – August 2024; May 2025 – August 2025*

- Refactored and migrated a Synthetic Aperture Radar (SAR) Automatic Target Recognition (ATR) framework from TensorFlow to MATLAB using the Deep Learning Toolbox, improving maintainability and streamlining deployment
- Developed and validated radar and geospatial utilities for LLA, ECEF, ENU, and UTM coordinate transformations and 3D visualization of beam patterns, iso-range spheres, iso-Doppler cones, velocity vectors, and angular geometry
- Built image-processing tools to suppress spurious artifacts and reconnect discontinuous regions in binary images

Harman International Novi, MI
System Test Engineer Intern | *Arduino, C#, HTML/CSS, PHP, SQL* *May 2023 – August 2023*

- Spearheaded the Harman Phone Management System, an internal platform for storing, securing, and managing engineering test devices, integrating an Arduino-controlled deadbolt with a web interface and administrative tooling
- Strengthened platform security by implementing HTTPS with OpenSSL-generated SSL certificates and PHP session-based authentication for secure user and database management

Projects

Secure AI Gateway | *Python, FastAPI, PostgreSQL, Redis, Kubernetes*

- Built an OpenAI-compatible LLM security gateway with API-key authentication, model authorization, distributed rate limiting, usage budgets, PII and prompt-injection controls, and least-privilege tool authorization
- Engineered PostgreSQL/Redis-backed authoritative state, hardened Docker/Kubernetes deployment, Prometheus/OpenTelemetry observability, and adversarial, resilience, security, and supply-chain verification in CI

Detecting Dirty Dish Culprits | *Python, YOLOv8, Roboflow, OpenCV*

- Built a YOLOv8-based computer vision pipeline to classify washing versus walking behavior in kitchen footage and flag likely dirty-dish drop-offs
- Annotated custom video data, added temporal continuity logic, and evaluated the detector with a 70/20/10 split, achieving 0.96 mAP50

Decrypting Blockchain: The Art of the Steal | *Cryptography, Blockchain, Security Analysis*

- Researched Bitcoin and Monero security/privacy tradeoffs, analyzing ring signatures, RingCT, stealth addresses, Dandelion++, traceability limits, and scam case studies

Bisdorf.org | *Astro, TypeScript, CSS, Cloudflare Pages*

- Built and deployed a responsive portfolio site with reusable Astro layouts, static project data, resume rendering/download support, project pages, and search indexing

SillyQL | *C++, Hash Tables, Binary Search Trees, Query Processing*

- Built a simplified SQL-style database engine using hash tables and binary search trees to support command-driven query processing and table joins

Network Security | *Python, Network Security, Protocol Analysis, Packet Handling*

- Implemented security-focused networking analyses with Python packet-handling tooling, evaluating protocol behavior, attacks, and defenses